

CONTACT

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ACADEMIC

Postdoctoral Research Fellow (*part-time*)

Aug 2025-now

CAREER

Department of Physics and Astronomy, University of Exeter | Exeter, UK

Postdoctoral Research Fellow (*incl. 9 months maternity leave*)

Apr 2022-Jul 2025

Department of Physics and Astronomy, University of Exeter | Exeter, UK

Postdoctoral Research Fellow

Sep 2019-Mar 2022

Department of Physics and Astronomy, University of Exeter | Exeter, UK

EDUCATION

PhD in Environmental Sciences

2015-2020

School of Environmental Sciences, University of East Anglia | Norwich, UK

Supervisors: Prof. Claire Reeves, Dr Paul Griffiths, Dr Marcus Köhler, Dr Mike Newland

Thesis: Impacts of C₁-C₃ alkyl nitrates on tropospheric ozone chemistry**MSc in Climate Change with Distinction**

2014-2015

School of Environmental Sciences, University of East Anglia | Norwich, UK

Supervisor: Prof. Claire Reeves

Thesis: Investigation of the relationship between tropospheric ozone production efficiency and carbon bond emissions

Specialist Diploma in Meteorology

2009-2014

Faculty of Geography, Lomonosov Moscow State University | Moscow, Russia

Supervisor: Prof. Alexander V. Kislov

Thesis: Climatically-induced variations of the Caspian Sea level over the last Millennium

PUBLICATIONS

26. **Zamyatina M.**, Hébrard E., Baskett H., Veillet R. et al. (in review).
Spatial variability of CH₄-to-CO conversion pathways in gas giant atmospheres: implications for chemical relaxation and 1D quench schemes.
25. Steinrueck M.E., Savel A.B., Christie D.A. et al. (incl. **Zamyatina M.**) (in review).
[Limb asymmetries on WASP-39b: A multi-GCM comparison of chemistry, clouds, and hazes.](#)
24. Christie D.A., Evans-Soma T.M., Mayne N.J., (incl. **Zamyatina M.**) (2026).
[Transport and thermochemical kinetics of alkali species in hot Jupiter atmospheres: implications for magnetic models.](#)
23. Venot O., Miguel Y., Baeyens R. et al. (incl. **Zamyatina M.**) (*section lead*; 2026).
[Probing exoplanetary chemistry with Ariel: scientific priorities and observational Strategies.](#)
22. Mukherjee S., Sing D.K., Fu G. et al. (incl. **Zamyatina M.**) (2026). *Science*.
[Cloudy mornings and clear evenings on a giant extrasolar world.](#)
21. Christie D.A., **Zamyatina M.**, Hébrard E., Evans-Soma T.M. et al. (2026). *MNRAS*.
[Benchmarking two chemical networks used in general circulation models of hot Jupiters.](#)
20. Radica M., Taylor J., Rotman Y. et al. (incl. **Zamyatina M.**) (2026). *AJ*.
[Super-solar metallicity and tentative evidence for photochemistry on WASP-96b from JWST and ground-based VLT transmission spectroscopy.](#)
19. Fairman C., Wakeford H.R., Claringbold A.B. et al. (incl. **Zamyatina M.**) (2026). *MNRAS*.
[BOWIE-ALIGN: Exploring degeneracies in the muted transmission spectrum of the aligned hot Jupiter NGTS-2b with NIRSpec/G395H.](#)
18. Claringbold A.B., Fisher C.E., Kirk J. et al. (incl. **Zamyatina M.**) (2026). *MNRAS*.
[BOWIE-ALIGN: Sub-solar C/O ratio and metallicity atmosphere of the misaligned hot Jupiter HAT-P-30b.](#)
17. Ahrer E-M., Fairman C., Kirk J. et al. (incl. **Zamyatina M.**) (2025). *MNRAS*.
[BOWIE-ALIGN: Weak spectral features in KELT-7b's JWST NIRSpec/G395H transmission spectrum imply a high cloud deck or a low-metallicity atmosphere.](#)
16. Fu G., Mukherjee S., Stevenson K.B. et al. (incl. **Zamyatina M.**) (2025). *ApJL*.
[Overcast mornings and clear evenings in hot Jupiter exoplanet atmospheres.](#)
15. Mak M.T., Sergeev D.E., Mayne N.J., **Zamyatina M.** et al. (2025). *MNRAS*.
[The impact of different haze types on the atmosphere and observations of hot Jupiters: 3D simulations of HD 189733b, HD 209458b and WASP-39b.](#)
14. Ahrer E-M., Gandhi S., Alderson L. et al. (incl. **Zamyatina M.**) (2025). *MNRAS*.
[Tracing the formation and migration history: molecular signatures in the atmosphere of misaligned hot Jupiter WASP-94Ab using JWST NIRSpec/G395H.](#)
13. Meech A., Claringbold A.B., Ahrer E-M. et al. (incl. **Zamyatina M.**) (2025). *MNRAS*.

- BOWIE-ALIGN: Sub-stellar metallicity and carbon depletion in the aligned TrES-4b with JWST NIRSpec transmission spectroscopy.
12. Kirk J., Ahrer E.-M., Claringbold A.B., **Zamyatina M.** et al. (2025). MNRAS.
BOWIE-ALIGN: JWST reveals hints of planetesimal accretion and complex sulphur chemistry in the atmosphere of the misaligned hot Jupiter WASP-15b.
 11. Penzlin A.B.T., Booth R.A., Kirk J. et al. (incl. **Zamyatina M.**) (2024). MNRAS.
BOWIE-ALIGN: How formation and migration histories of giant planets impact atmospheric compositions.
 10. Kirk J., Ahrer E.-M., Penzlin A.B.T. et al. (incl. **Zamyatina M.**) (2024). RASTI.
BOWIE-ALIGN: A JWST comparative survey of aligned versus misaligned hot Jupiters to test the dependence of atmospheric composition on migration history.
 9. Espinoza N., Steinrueck M., Kirk J. et al. (incl. **Zamyatina M.**) (2024). Nature.
Inhomogeneous terminators on the exoplanet WASP-39 b.
 8. Christie D.A., Mayne N.J., **Zamyatina M.**, Baskett H. et al. (2024). MNRAS.
Longitudinal filtering, sponge layers, and equatorial jet formation in a general circulation model of gaseous exoplanets.
 7. **Zamyatina M.**, Christie D.A., Hébrard E., Mayne N.J. et al. (2024). MNRAS.
Quenching-driven equatorial depletion and limb asymmetries in hot Jupiter atmospheres: WASP-96b example.
 6. Taylor J., Radica M., Welbanks L. et al. (incl. **Zamyatina M.**) (2023). MNRAS.
Awesome SOSS: atmospheric characterisation of WASP-96b using the JWST early release observations.
 5. Radica M., Welbanks L., Espinoza N. et al. (incl. **Zamyatina M.**) (2023). MNRAS.
Awesome SOSS: transmission spectroscopy of WASP-96b with NIRISS/SOSS.
 4. **Zamyatina M.**, Hébrard E., Drummond B., Mayne N.J. et al. (2023). MNRAS.
Observability of signatures of transport-induced chemistry in clear atmospheres of hot gas giant exoplanets.
 3. Christie D.A., Lee E.K.H., Innes H. et al. (incl. **Zamyatina M.**) (2022). PSJ.
CAMEMBERT: A Mini-Neptunes GCM Intercomparison, Protocol Version 1.0. A CUISINES Model Intercomparison Project.
 2. Ridgway R.J., **Zamyatina M.**, Mayne N.J., Manners J. et al. (2023). MNRAS.
3D modelling of the impact of stellar activity on tidally locked terrestrial exoplanets: atmospheric composition and habitability.
 1. Braam M., Palmer P.I., Decin L., Ridgway R.J., **Zamyatina M.** et al. (2022). MNRAS.
Lightning-induced chemistry on tidally-locked Earth-like exoplanets.

INVITED TALKS

- | | |
|----------|---|
| May 2026 | Are hot Jupiters growing diamonds?
University of Bristol (chemistry seminar) Bristol, UK |
| Mar 2026 | Out of equilibrium in 3D: disequilibrium chemistry in gas giants (<i>postponed due to illness</i>)
Queen Mary University of London (astronomy seminar) London, UK |
| Jul 2025 | Atmospheric chemistry of solar and extrasolar gas giants (<i>review</i>)
Exoclimes VII conference Montreal, Canada |
| Mar 2024 | Overview of the Met Office Unified Model configuration for hot Jupiter atmospheres
International Space Science Institute (ISSI) workshop Bern, Switzerland |
| Feb 2024 | Quenching-driven equatorial depletion and limb asymmetries in WASP-96b's atmosphere
University of Bristol (astronomy seminar) Bristol, UK |
| Feb 2023 | Atmospheric dynamics and chemistry on exoplanets
University of Queensland (astronomy seminar) Brisbane, Australia
University of Southern Queensland (exoplanet seminar) Toowoomba, Australia
University of New South Wales (astronomy seminar) Sydney, Australia |
| Nov 2022 | Observability of signatures of wind-driven chemistry in atmospheres of hot gas giants
Ludwig Maximilian University (exoplanet group seminar) Munich, Germany
Celebrating JWST's first six months of exoplanet data workshop Ringberg castle, Germany |
| Oct 2022 | Modelling chemistry of hot Jupiter atmospheres with the Met Office Unified Model
Met Office Exeter, UK |
| Feb 2022 | Transport-induced quenching shapes transmission spectra of warm and hot Jupiters
University of Warwick (astronomy seminar) virtual |

CONTRIBUTED TALKS

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| Apr 2026 | Spatial variability of CH ₄ -to-CO conversion pathways in gas giant atmospheres
Ariel chemistry working group meeting virtual |
| Sep 2023 | Metallicity masquerade: how to use quenching to distinguish between different planet metallicities
University of Bristol (BOWIE meeting) Bristol, UK |
| June 2021 | Overview of the Met Office Unified Model adapted to simulate exoplanetary atmospheres
Ariel consortium meeting virtual |

	Apr, Sep 2021	3D simulations of warm and hot Jupiter atmospheres: the role of 3D mixing in shaping CH ₄ -to-CO conversion pathways EPSC conference virtual UKEXOM conference virtual University of Exeter (astronomy seminar) Exeter, UK	
	Mar, Apr, Jun 2019	Impact of C ₁ -C ₃ alkyl nitrate chemistry on tropospheric ozone: box and global model perspectives University of Exeter (XCS seminar) Exeter, UK EGU conference Vienna, Austria University of East Anglia (AMB seminar) Norwich, UK	
	Apr 2017	Adding new chemistry into UM-UKCA Cambridge-EnvEast doctoral alliance symposium Cambridge, UK	
	Sep 2012	Assessment of climatological potential of transboundary air pollution transport in Eastern Siberia and the Russian Far East Air quality management at urban, regional and global scales 4th international symposium/IUAPPA regional conference Istanbul, Turkey	
POSTERS	Jul 2025	CH ₄ -CO conversion pathways in hot Jupiter atmospheres Exoclines VII conference Montreal, Canada	
	Apr, Jun 2024	Quenching-driven equatorial depletion and limb asymmetries in WASP-96b's atmosphere UKEXOM conference Birmingham, UK Exoplanets 5 conference Leiden, Netherlands	
	Sep 2022	Applying known chemical kinetics data to model atmospheres of extrasolar planets iCACGP-IGAC conference Manchester, UK	
	Sep 2021	Local and global impacts of C ₁ -C ₃ alkyl nitrate photochemistry and emissions on tropospheric ozone IGAC conference virtual	
	Sep 2018	Impact of alkyl nitrate chemistry on tropospheric ozone iCACGP-IGAC conference Takamatsu, Japan	
	Mar, Apr 2018	Impact of C ₁ -C ₅ alkyl nitrate chemistry on tropospheric ozone - a box modelling study Cambridge-EnvEast doctoral alliance symposium Cambridge, UK EGU conference Vienna, Austria	
AWARDED FUNDING	2023-2026	Co-I DIRAC-3 operations 2023-2026 (PI: Nathan Mayne; for RSE)	26,302.66£
	2023	Above & Beyond Award	
	2022	EPSRC vacation internship (for 3 interns)	12,893.55£
	2022	Jackson-Grime-Davies (JGD) research internship (for 1 intern)	2,428.71£
	2021	IGAC Early Career Scientist poster prize & travel grant	1,227.70£
	2015-2019	Lord Zuckerman studentship	112,269.50£
	2014-2015	Simon Wharmby postgraduate scholarship	3,000.00£
	2012	World Meteorological Organization travel grant	1,154.10£
AWARDED OBSERVING TIME	Mar 2026 (C5)	Cloudy with a chance of disequilibrium chemistry: A comprehensive view of silicate clouds in a hot Jupiter with a MIRI/LRS phase curve ■ Co-I JWST GO-11301 (PI: Jared Splinter)	69.5 hours
	Feb 2024 (C3)	JWST's exoplanet grand tour spectroscopic survey ■ Co-I HST GO-17612 (PI: David Sing) ■ Co-I JWST GO-5924 (PI: David Sing)	24 orbits 125.70 hours
	Feb 2024 (C3)	Starspots, hazes, and disequilibrium chemistry: a deep dive into the atmosphere of HAT-P-18b ■ Co-I JWST GO-5844 (PI: Michael Radica)	16.40 hours
	May 2023 (C2)	Putting it all together: Dynamics and chemistry probed through transmission spectroscopy of a cloud-free exoplanet ■ Co-I JWST GO-4082 (PI: Michael Radica, Co-PI: Jake Taylor)	6.69 hours
	May 2023 (C2)	Hot Jupiter atmospheric forecast: are mornings cloudier than evenings in other worlds? ■ Co-I JWST GO-3969 (PI: Nestor Espinoza, Co-PI: Diana Powell)	61.53 hours
	May 2023 (C2)	Does atmospheric composition actually trace formation? Observing aligned vs misaligned hot Jupiters as a testbed ■ Co-I JWST GO-3838 (PI: James Kirk Co-PI: Eva-Maria Ahrer)	49.21 hours
	May 2023 (C2)	Testing the C/O ratio prediction for hot Jupiters from disk-free migration ■ Co-I JWST GO-3154 (PI: Eva-Maria Ahrer)	10.36 hours
SUPERVISION		Primary supervisor and co-supervisor. Students who went on to do a PhD are marked with *.	

		PhD supervision (2)
TEACHING	Sep 2024-now	Harry Baskett Thesis: TBD Co-supervisors: Dr. E. Hebrard, Prof. N. J. Mayne
	Nov 2020-May 2023	Robert J. Ridgway Thesis: Simulating the impact of stellar flares on the climate and habitability of terrestrial Earth-like exoplanets Co-supervisors: Prof. N. J. Mayne, Prof. F. H. Lambert, Dr. J. Manners
	Jun-Aug 2022	Undergraduate and summer internship supervision (4) EPSRC-funded: Harry Baskett*, Ben Moore* , James McDermott* ; JGD-funded: Graig Lils Project: 3D modelling of hot Saturn atmospheric chemistry
TEACHING	Jul 2023	Module leader Module: No place like home: placing Earth in its geological and astronomical contexts International sustainability summer school University of Exeter, Exeter, UK
	Jul 2022, Jul 2023	Lecturer Module: No place like home: placing Earth in its geological and astronomical contexts International sustainability summer school University of Exeter, Exeter, UK
	Sep 2021-Feb 2022	Associate Tutor Modules: Experimental science, Frontiers in science University of Exeter Exeter, UK
	Jan 2018	Instructor Module: Introduction to Python in Environmental Sciences University of East Anglia Norwich, UK
	2015-2018	Associate Tutor Modules: Numerical skills for scientists, Atmospheric chemistry and global change, Atmospheric composition (measurements and modelling), Atmosphere & oceans I University of East Anglia Norwich, UK
ACADEMIC COMMUNITY		Professional memberships
ACADEMIC COMMUNITY	Sep 2026-Aug 2029	STFC DiRAC Resource Allocation Committee: astronomy and cosmology sub-panel (member)
	2022-now	Ariel Space Mission chemistry working group (member)
		Organisation of scientific meetings
	26-30 Jun 2023	Exoclines VI conference (LOC member, session chair) ~200 attendees University of Exeter Exeter, UK
	22-24 Jun 2023	ExoSLAM school (LOC member) ~50 attendees University of Exeter Exeter, UK
	5-6 Dec 2022	BOWIE meeting (co-organiser) 17 attendees University of Exeter Exeter, UK
	Sep 2017-Jun 2018	Atmospheric and Marine Biogeochemistry (AMB) seminars (co-organiser) ~20 attendees University of East Anglia Norwich, UK
		Reviewing
		Journals: The Astrophysical Journal (3), Monthly Notices of the Royal Astronomical Society (1) Proposals: JWST Cycle 3 TAC external expert, JWST Cycle 4 TAC panelist
OUTREACH		
OUTREACH	Jul 2024	Joint press release about Espinoza et al. (2024) paper : Research confirms that distant world's eternal sunrise and sunset are not alike
	Sep 2023	Expert scientist at the Climate Exhibition (part of the British Science Festival)
	Nov 2015-Jun 2019	Maintainer of @AtmosChemUEA Twitter account
VOCATIONAL EXPERIENCE		
VOCATIONAL EXPERIENCE	Aug-Sep 2013	Weather Forecaster Sheremetyevo International Airport Moscow, Russia
	Jun-Jul 2013	Technician Department of Actinometry, Meteorological Observatory Lomonosov Moscow State University Moscow, Russia
VOCATIONAL TRAINING		
VOCATIONAL TRAINING	Sep 2023	Belbin training
	Mar 2023	Leadership training
	Dec 2022	Interview training
	Sep 2021	Learning and Teaching in Higher Education (LTHE) Unit 1
	Mar 2020	JWST proposal planning workshop
	Apr 2016	NAME workshop
VOCATIONAL TRAINING	Jan 2016	Introduction to UKCA

Dec 2015 Introduction to Unified Model
Nov 2015 [Introduction to Atmospheric Science](#)
2015-2019 [EnvEast Doctoral Training Programme](#)